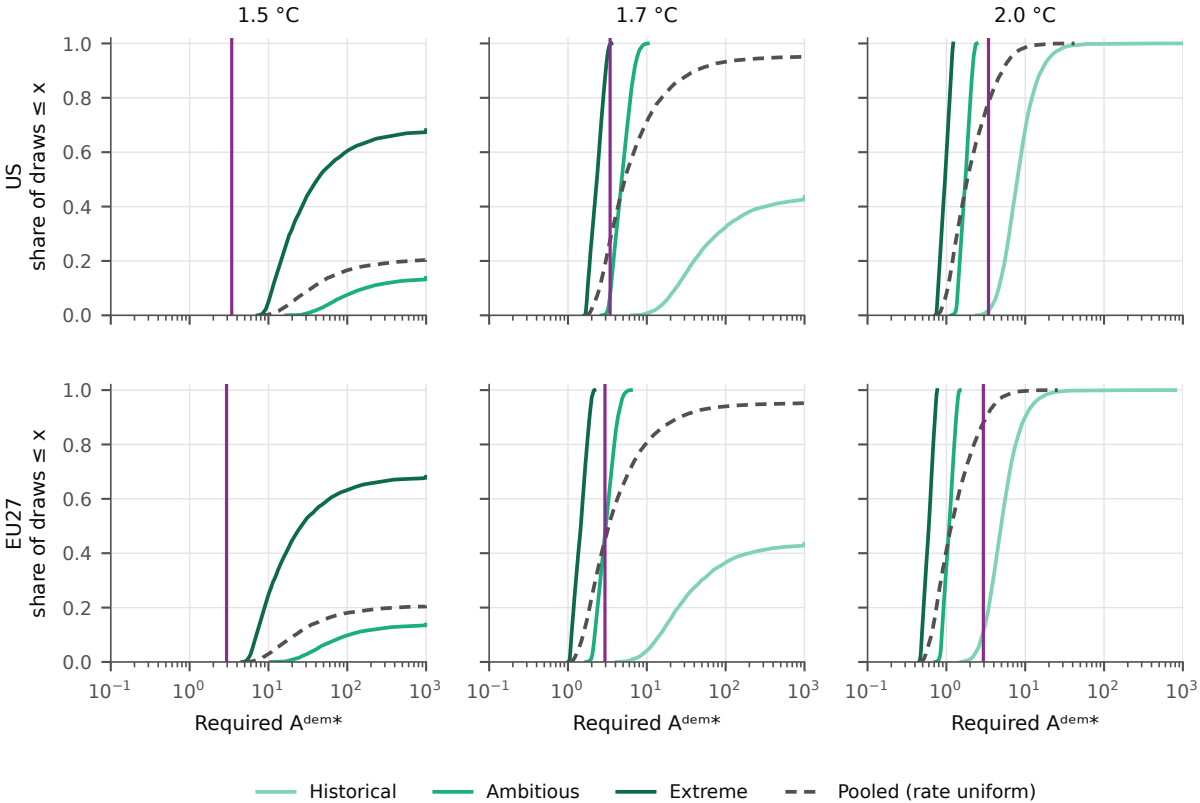S2. Monte Carlo distribution of required A^{dem*} (path frame, 10,000 draws): curves that end below 1 mean the floor fails in the remaining draws; purple line = observed A^{dem}